

AERO50001 Aerodynamics 2 Problem Set #4

Extra practice: Oblique shocks

4.1 Shock polar

★☆☆ 5-10 mins

Use the shock polar to answer the following questions (estimate to about 0.5°).

- (a) What is the deflection angle that will make the flow behind a shock at $M = 2.4$ sonic?
 - (b) What is the maximum deflection angle for which a flow at $M = 1.6$ will remain attached?
 - (c) What deflection angle will produce a shock with wave angle of 60° at $M = 1.9$?
 - (d) What will be the the shock wave angle when a flow at Mach 3 is deflected by 12° ?
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(a) 28.5°

(b) 14.6°

(c) 20.5°

(d) $29.2^\circ \approx 29.5^\circ$

4.2 Oblique shock

★★☆ 15-25 mins

Consider a supersonic flow ($\gamma = 1.4$) with $M_1 = 2.0$, $p_1 = 1$ atm and $T_1 = 288$ K. This flow is deflected at a compression corner of 20° .

- (a) Determine the shock wave angle β .
- (b) Calculate the Mach number M_2 behind the resulting oblique shock wave.
- (c) Calculate the pressure p_2 behind the oblique shock wave.
- (d) Calculate the temperature T_2 behind the oblique shock wave.
- (e) Calculate the stagnation pressure p_{02} behind the oblique shock wave.
- (f) Calculate stagnation temperature T_{02} behind the oblique shock wave.

$$(a) \beta = 53.5^\circ$$

$$(b) \because M_2^2 \sin^2(\beta - \theta) = \frac{2 + (\gamma - 1) M_1^2 \sin^2 \beta}{2\gamma M_1^2 \sin^2 \beta - (\gamma - 1)}$$

$$\therefore M_2 = 1.207$$

$$(c) p_2 = p_1 \left(\frac{2\gamma M_1^2 \sin^2 \beta}{\gamma + 1} - \frac{\gamma - 1}{\gamma + 1} \right) = 2.8468 \text{ atm}$$

$$(d) T_2 = T_1 \left[\frac{2 + (\gamma - 1) M_1^2 \sin^2 \beta}{2\gamma M_1^2 \sin^2 \beta - (\gamma - 1)} \right] \frac{2\gamma M_1^2 \sin^2 \beta - (\gamma - 1)}{(\gamma + 1)^2 M_1^2 \sin^2 \beta}$$
$$= 401.27 \text{ K}$$

$$(e) p_{02} = p_2 \left(1 + \frac{\gamma - 1}{2} M_2^2 \right)^{\frac{\gamma}{\gamma - 1}} = 6.9717 \text{ atm}$$

$$(f) T_{02} = T_2 \left(1 + \frac{\gamma - 1}{2} M_2^2 \right) = 518.19 \text{ K}$$